

Jin Huang (Hugo) Research Engineer @ Huawei R&D UK

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If any information seem outdated, latest version of this resume is available at hugohuang.com
[ORCID](#) [Google Scholar](#) [OpenReview](#)

Publication

- 2023 [BEV@DC: Bird’s-Eye View Assisted Training for Depth Completion](#)
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)
Wending Zhou, Xu Yan, Yinghong Liao, Yuankai Lin, [Jin Huang](#), Gangming Zhao, Shuguang Cui, Zhen Li
Computer Vision (Depth Completion)
- 2025 [LLM Shots: Best Fired at System or User Prompts?](#)
WWW ’25: Companion Proceedings of the ACM on Web Conference 2025
Umut Halil, Jin Huang, Damien Graux, Jeff Z Pan
Natural Language Processing (In-Context Learning, Prompt Engineering)

Current Position

Research Engineer @ Huawei (2012 Lab / Central Software Institute / RAI Lab)

Nov. 2024 – Current

Engineering & Deployments:

- **2025 Annual Performance Review: Excellent** (top 10%).
- Received **Star of Edinburgh Award** for my contribution to Huawei’s LLM Agent System, [openJiuwen](#). Led development for Knowledge Base Retrieval, Graph-based Agent Memory and the internal demo for Context Engine.
- Top contributor to the [openJiuwen](#) ecosystem, helped with code reviews & urgent bug fixes, only overseas contributor in top-20. In charge of code review & approval for all pull requests from Edinburgh.
- Led development for multiple features to close-source Jiuwen, deployed to Huawei Cloud and a Top-4 Bank in China.
- Helped to establish internal collaboration projects with other teams within Edinburgh Research Center & with teams from China HQ. Also was admin to most servers within the lab, maintaining internal model / database services.

Research-Related:

- **Program Committee Member**, PromptEng workshop – to be co-located with the ACM Web Conference 2026.
- **Challenge Designer**, CCF Big Data & Computing Intelligence Contest 2026 – hosted by [China Computer Federation](#).
- Helped with [openJiuwen-deepsearch](#), claimed the [BrowseComp-Plus](#) leaderboard in 2025 with GPT-5-mini.
- Collaborated with programming language department on code conversion project to aid Huawei customers convert iOS or Android application code base to work on HarmonyOS, building code graph to reduce noise in agentic exploration.
- Explored workflow generation with automated feedback, improved generation accuracy of ComfyUI workflows by 25.8%.
- Designed an interactive Program-of-Thoughts approach, enhancing LLM agents with feedbacks generated from code execution results, improved LLM agents’ performance on complex STEM & Math reasoning tasks, reached 74% accuracy on TheoremQA dataset with 14 & 32B Qwen2.5 models without fine-tuning. (Back in 2024)

Open Source Maintainer @ [Farama Foundation](#)

Jun. 2026 – Current

- Maintainer of [HighwayEnv](#) and a [core team](#) member; also help to maintain other Farama projects such as [Gymnasium](#) (OpenAI Gym) and [PettingZoo](#) (basically multi-agent Gym).
- We maintain many of the world’s open-source RL tools: 850+ GitHub contributors, 450M+ installations, and 35k+ public repositories on GitHub depending on us :-)

Education

Master’s Degree	MSc Artificial Intelligence	University of Edinburgh	Sep. 2023 – Aug. 2024
Bachelor’s Degree	BSc Computer Science	Cardiff University	Sep. 2020 – Jun. 2023

Open Source Contributions Outside of Work

HighwayEnv (3.3k stars on GitHub) <i>RL Driving Environment</i> Maintainer	A collection of environments for autonomous driving and tactical decision-making tasks, part of the Farama Foundation ecosystem. Used widely for research in RL for driving, planning, and multi-agent interaction. <i>Maintains HighwayEnv and other Farama projects.</i>
pip Contributor	The package installer for Python. <i>Added documentation & regression tests, marked official support for self-referential extras.</i>

ViZDoom

(390k downloads on [PyPI](#))

RL & CV Environment

Top Contributor

Cited in thousands of papers, ViZDoom allows developing AI that play Doom using only visual info. Intended for research in Computer Vision & Reinforcement Learning.

I built automated pipeline for Python binding's type-hinting support, in-engine object categorization (Semantic Segmentation) and improved / fixed various stuffs like seeding behavior, building process (GitHub Action & CMake), documentation & examples, etc.

Stable Baselines3

(24M downloads on [PyPI](#))

RL Framework

Contributor

Stable Baselines3 (SB3) is a set of reliable implementations of Reinforcement Learning algorithms in PyTorch. It is the next major version of Stable Baselines. One of the most popular Reinforcement Learning frameworks.

Enhancement for rollout buffers, reduced memory usage (in SB3 & SBX - JAX version).

GitCode API

(30k downloads on [PyPI](#))

Python SDK for GitCode

Maintainer

An easy-to-use Python SDK / CLI tool / MCP Server (with MCPB support) for [GitCode](#), second largest Chinese Git platform, built with user-centric API design. Since its creation, many projects from me and colleagues at Huawei have started to integrate it due to the ease of use, good typing design, documentation coverage and lightweight nature. It is now the most downloaded GitCode tool.

SB3 Extra Buffers

(11k downloads on [PyPI](#))

RAM-Saving Tool for RL

Author

A collection of extra Stable Baselines3 buffer classes. Reducing memory usage drastically with minimal overhead via vectorized implementation of lossless compression algorithms. Experiments conducted with Atari games show > 90% memory save with negligible latency. Project starred by maintainers of SB3 & OpenAI Gym and featured in SB3 documentation.

SegDoom

(Master's Thesis Project)

Play Doom with RL & CV

Author

Enhancing Reinforcement Learning in 3D Environments through Semantic Segmentation: A Case Study in ViZDoom explored the potential of applying Semantic Segmentation (SS) to 3D video game environment and analyzed issues in previous literature. Experiments conducted in ViZDoom showed significant improvement with SS mask as augmentation to RGB input, and decrease in memory usage was observed if SS is used as a replacement to RGB (see sb3-extra-buffers). Novel visualization methods were developed to evaluate RL agents' suitability for data collection. Custom Semantic Segmentation dataset was created.

STD Raytracer

(Personal Project)

Ray-tracer / Path-tracer

Author

A ray-traced / path-traced renderer with support for custom models & textures, built from scratch with only C++ 17 standard library and nlohmann's json parser. Multi-threaded with bounding volume hierarchy acceleration, alternate frame rendering option, detailed logs and progress bar for user-friendliness. Support loading simple animation from Blender. An early version was submitted as Computer Graphics Rendering coursework at University of Edinburgh and got an excellent grade of 97%.

Past Internships

Backend Developer

SenseTime

Designed a tree-based RAG system for DevOps manuals, it preserves the structured nature of source materials and made use of keywords & synonyms to further provide alignment. Outperform official RAG implementation in SenseNova (SenseTime's LLM) at the time.

Jul. 2023 – Aug. 2023
Shenzhen, China

Research Assistant

Cardiff University

Developed a dataset creation tool in Unity Engine for synthesizing image sequences with mirror segmentation.

Jun. 2022 – Aug. 2022
Cardiff, UK

Past Voluntary Works

CS Tutor

SCIE (High School)

Worked as Computer Science tutor for GCSE students in my high school, Shenzhen College of International Education. All students that entered with C/D grade improved to above B in IGCSE Exam.

Sep. 2018 – Jun. 2019
Shenzhen, China

Skills

General Workplace Skill

AI R&D Specific Skill

Field of Research

Programming Language

Scientific Computing

Hosting Services/Demos

Vector/Graph Database

Agent Development

Open Source Projects

Communication, Teamwork, Translation (Between English and Chinese), Presentation
Bridging gaps between scientists' and engineers' mindsets during meetings & discussions
Natural Language Processing, Reinforcement Learning, Computer Vision
Python, C++, C#, Java, MATLAB, ACS, etc.
NumPy, Numba, PyTorch, Sci-kit Learn, Gensim, FAISS, etc.
Docker, Podman, Flask, FastAPI, Gradio, ECharts, Vis.js, etc.
Milvus, Chroma, Qdrant, Nebula, Neo4j, LadybugDB, Kuzu, SQLite-Vec, etc.
Tool & Resources (MCP, OpenAI tool), SDK (LangChain/Graph, openJiuwen, AutoGen)
GitHub Actions, Sphinx Documentation (Read the Docs), PyPI, writing pyproject.toml.